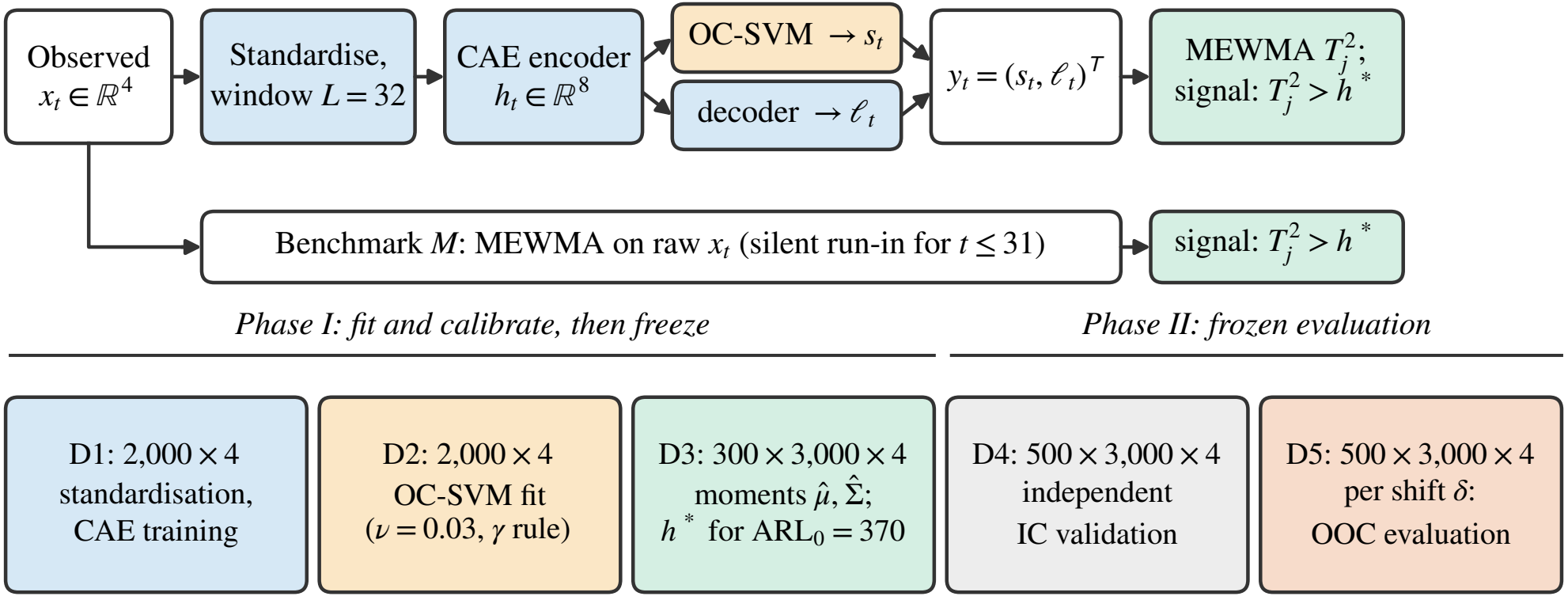
Observed
$x_t \in \mathbb{R}^4$
Standardise,
window $L = 32$
CAE encoder
$h_t \in \mathbb{R}^8$
OC-SVM $\rightarrow s_t$
decoder $\rightarrow \ell_t$
$y_t = (s_t, \ell_t)^\top$
MEWMA T_j^2;
signal: $T_j^2 > h^*$
Benchmark M: MEWMA on raw x_t (silent run-in for $t \leq 31$)
signal: $T_j^2 > h^*$
Phase I: fit and calibrate, then freeze
Phase II: frozen evaluation
D1: 2,000 × 4
standardisation,
CAE training
D2: 2,000 × 4
OC-SVM fit
($\nu = 0.03$, γ rule)
D3: 300 × 3,000 × 4
moments $\hat{\mu}, \hat{\Sigma}$;
h^* for $\mathrm{ARL}_0 = 370$
D4: 500 × 3,000 × 4
independent
IC validation
D5: 500 × 3,000 × 4
per shift δ:
OOC evaluation